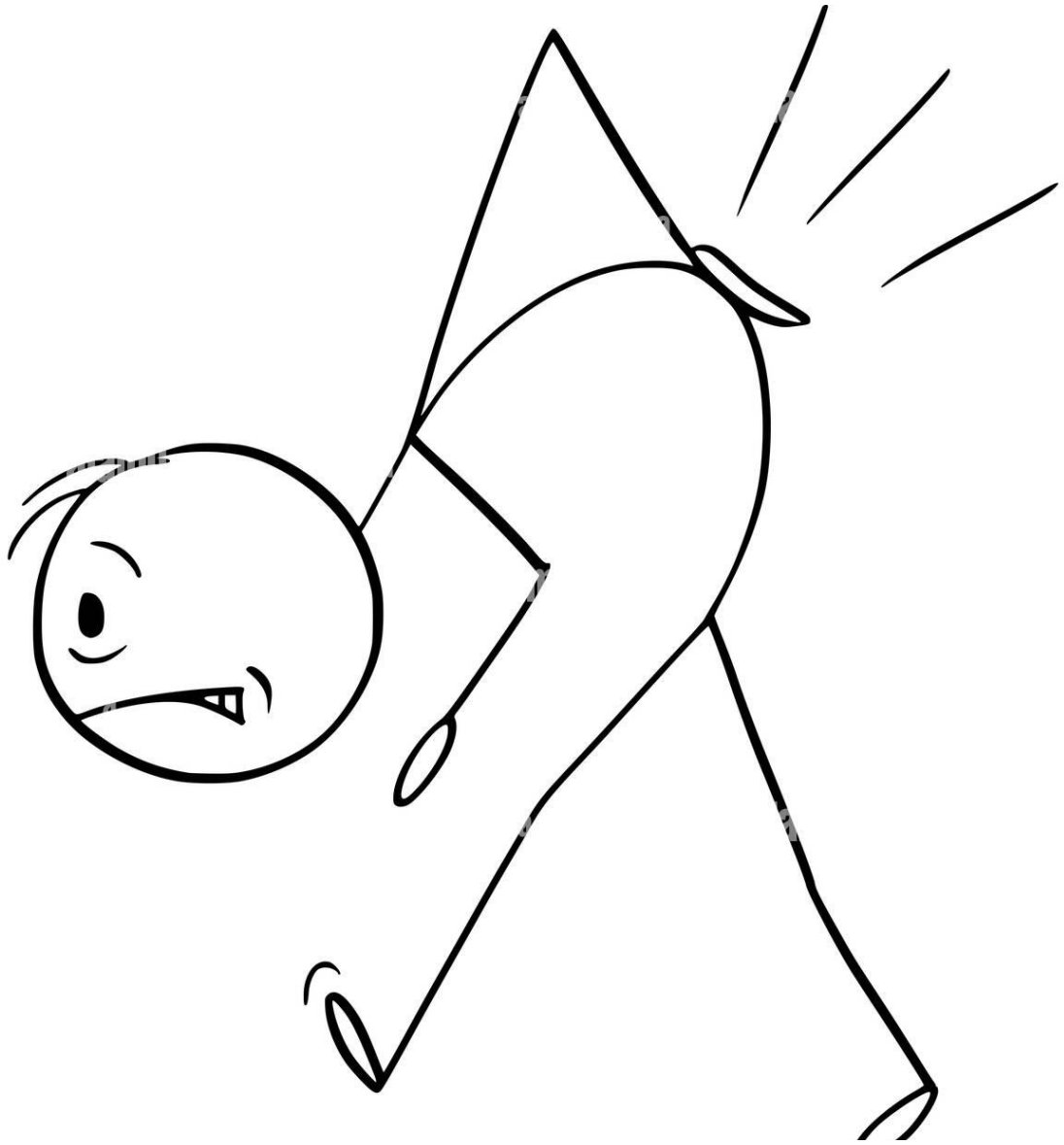
Communication in Pain Management PT5000

Lucy Metcalfe
Specialist Physiotherapist
Associate Lecturer

Learning Outcomes:

1. Understand the influencing factors on chronic pain (biopsychosocial model)
2. Describe the common treatment approaches in psychology including ACT and CBT
3. Explore the effective communication with patients with chronic pain

Why we Hurt



Tame the Beast

What is Pain?

Theory:

- “An unpleasant **sensory** and **emotional experience** associated with **actual** or **potential tissue damage**, or described in terms of such damage.” - IASP
- Pain is a subjective experience dependent on the self report of the individual

Reality:

- Experience produced by the brain
- Brain receives 1 000 000s of pieces of information at every second
- Result of the nervous and immune system interactions

Chronic Pain

- Chronic pain ➜ pain that persist beyond normal tissue healing time, which is assumed to be **3 months**
- In the UK, it is believed 20% of people suffer with chronic pain
- Chronic pain has a large financial impact, estimated at £12 billion in the UK in 2010, and disproportionately affects older people and those living in deprivation or isolation.

Common thoughts

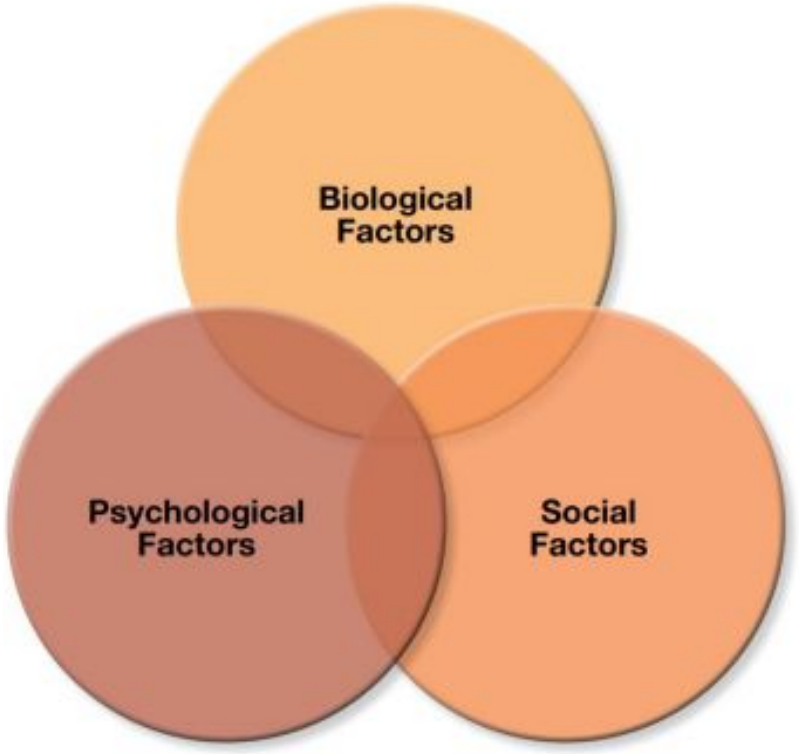
- Pain and nociception - are these the same?
- Varied theories - one theory is not conclusive
    - **Extensive literature contents that people's pain experience is multidimensional (BIOPSYCHOSOCIAL)**

Lorimer's Pain Story

- TEDxAdelaide - Lorimer Moseley - Why Things Hurt
- https://www.youtube.com/watch?v=gwd-wLdIHjs&t=1s

Biopsychosocial Model

- The biopsychosocial model was proposed by George Engel (1977) who voiced concern over the narrowness of the biomedical model based on his experiences with patients.



Influential Factors In Pain Experience

Psychological Factors

- Pain Cognitions and Negative thoughts
- Catastrophizing.
- Hurt versus Harm.
- Answer-Seeking.
- Pain Self-efficacy.

Social

- Solicitous Significant Other
- Punishing Responses.

Behavioural

Passive Coping: Guarding and Resting/Under-activity

Active Coping: Exercise and Over-activity.

Other Coping: Pacing

Yellow Flags

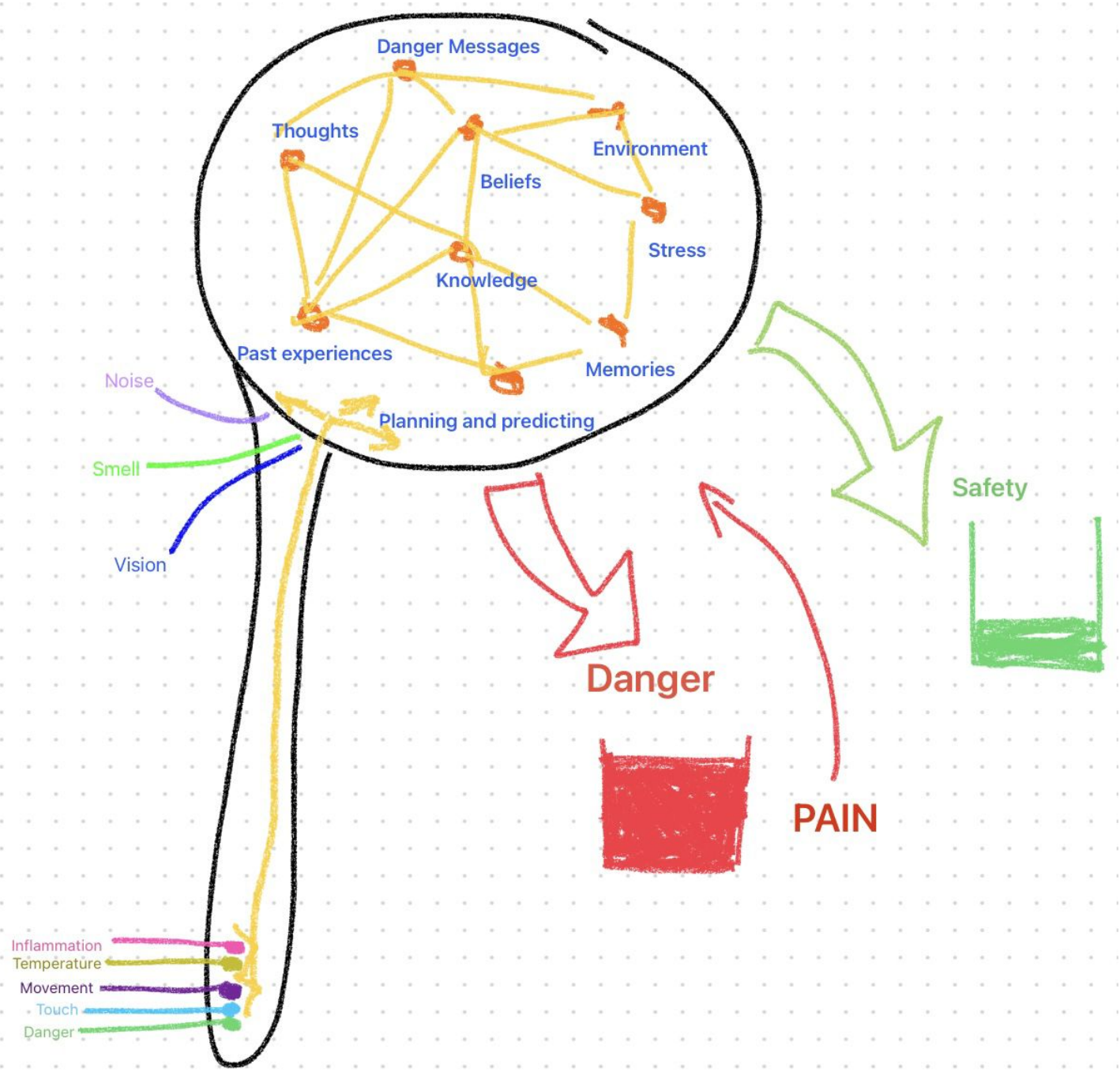
- Beliefs, appraisals and judgements
- Unhelpful beliefs about pain
- Expectations of poor treatment outcomes
- Pain behaviour (including pain and coping strategies)
- Avoidance of activities due to expectations of pain or reinjury
- Over reliance on passive treatments



Examples:

- Catastrophizing
- Finding painful experiences unbearable - reporting disproportionate pain
- Having unhelpful beliefs about pain and work
- Fear of movement or re injury
- Expecting other people or interventions to solve the program
- Multiple visits to various practitioners with NO improvement

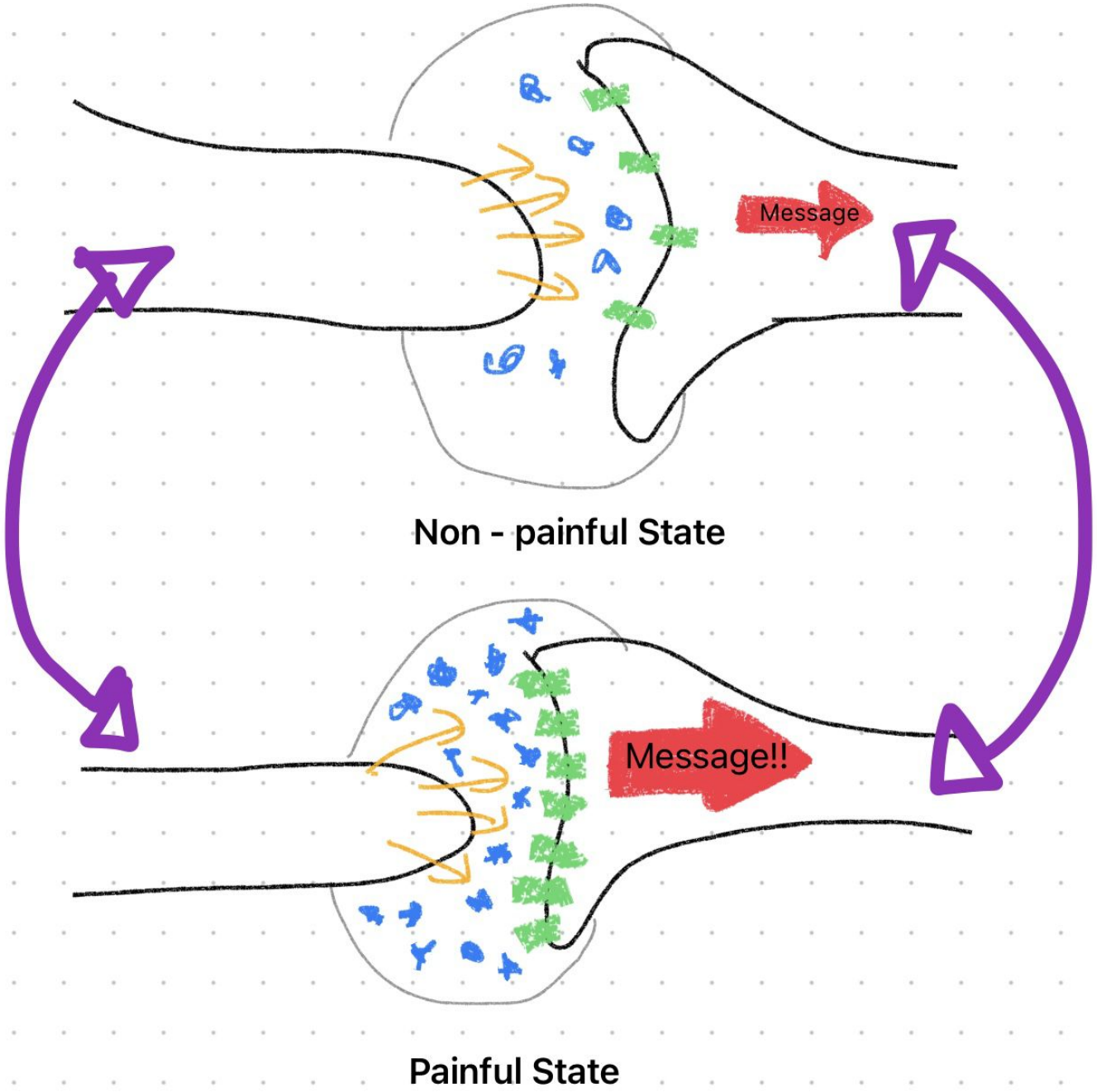
Danger vs Safety



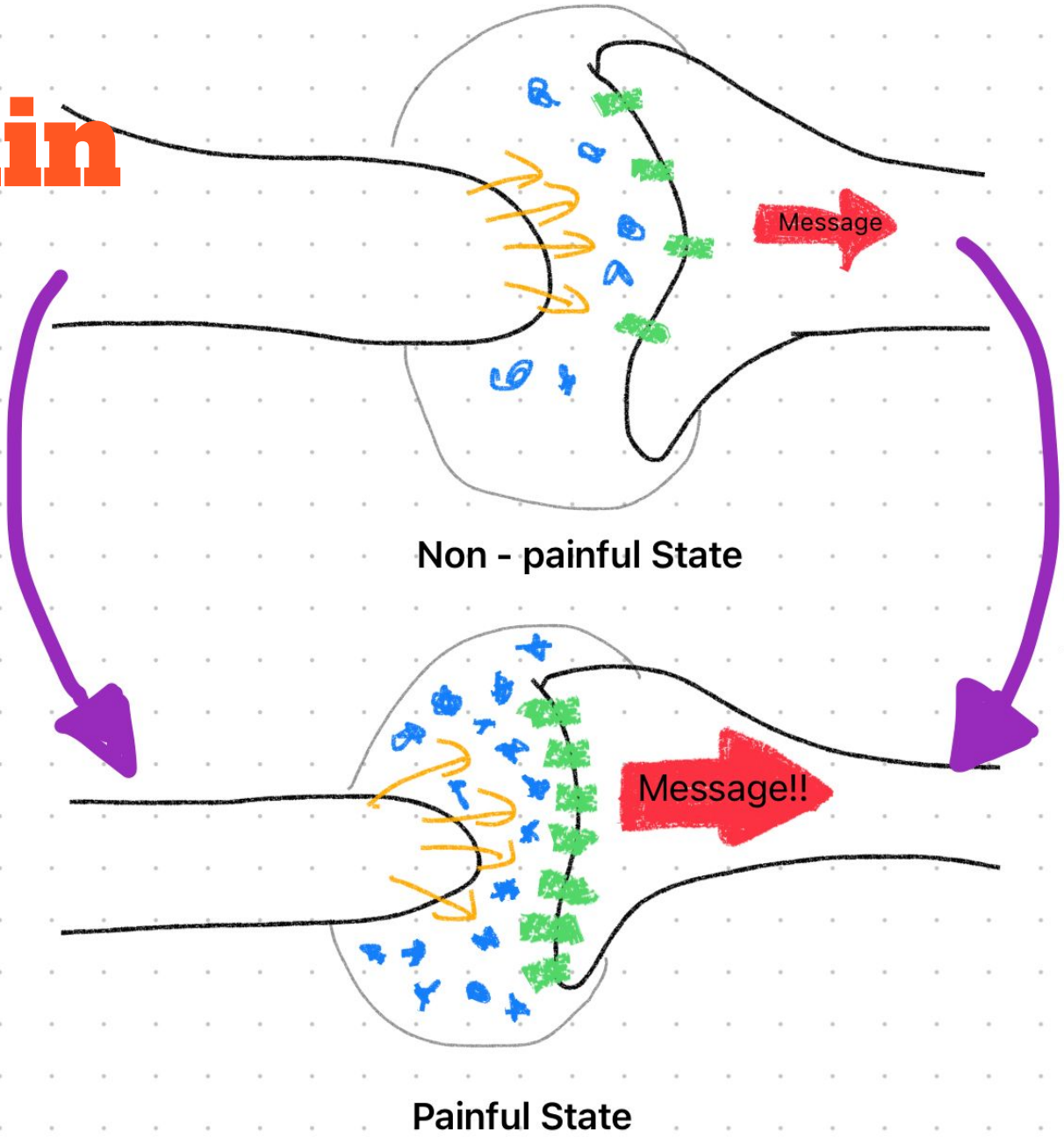
Normal Pain Response

Peripheral Sensitisation —>
Normal response to injury to increase inflammation for healing

= *Improved efficiency…*
Temporarily



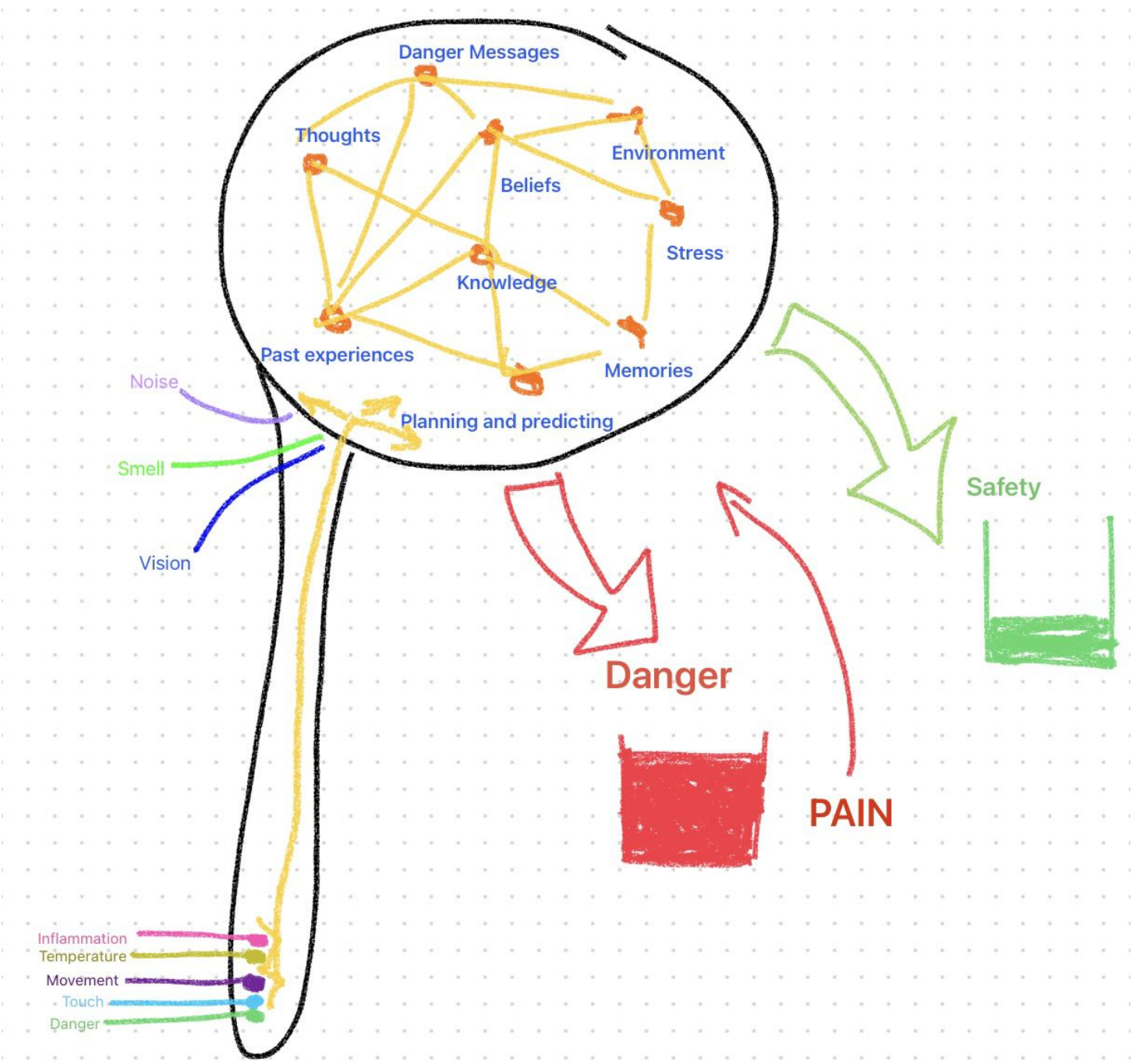
Chronic Pain State



Heightened immune system:
- sickness
- poor sleep
- poor diet
- High stress

Sensitised Nervous System:
- fear
- stress
- poor social network
- opioids (> 3 days)
- poor beliefs
- Guarded movement
- lack of endorphins: depression, lack of joyful activities etc

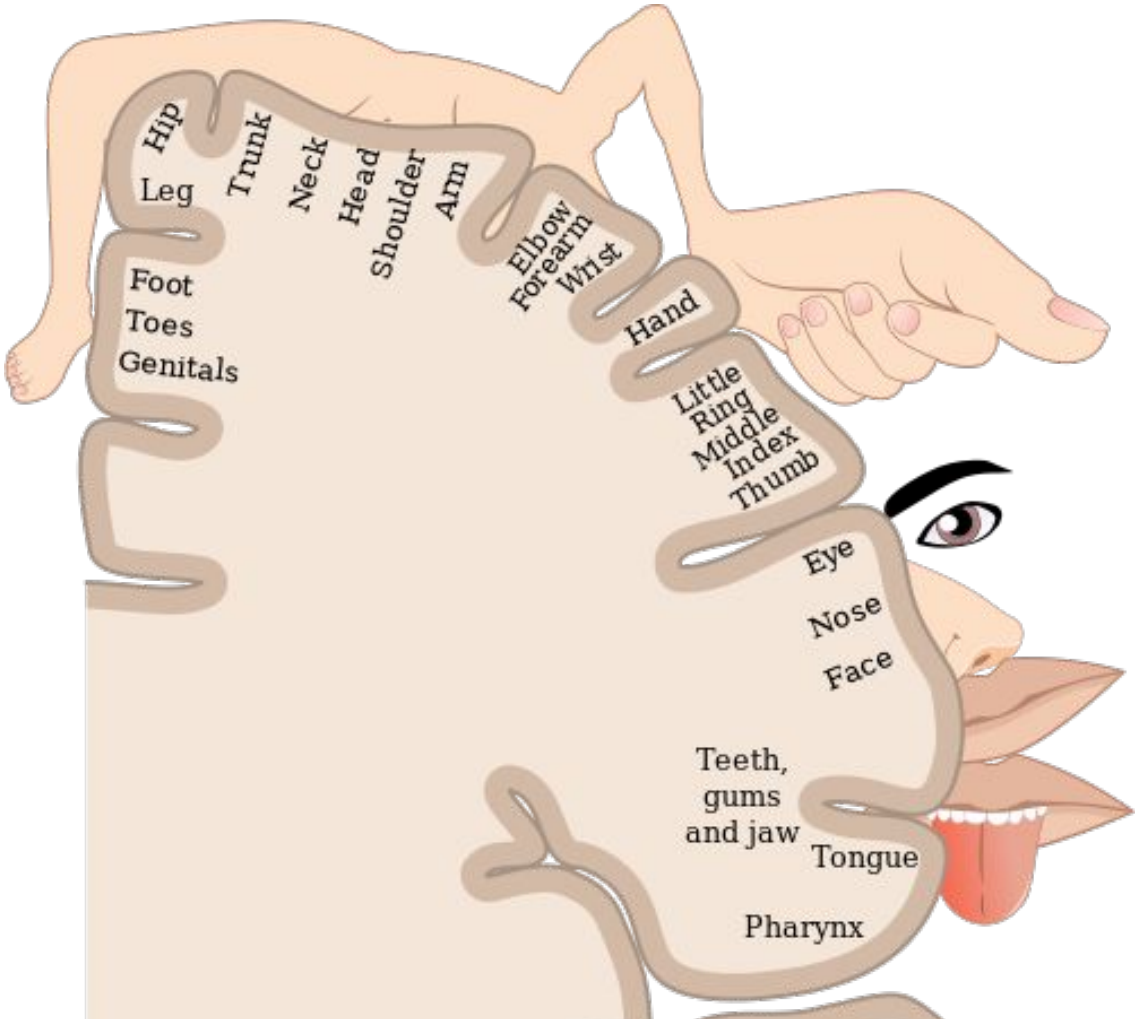
More Danger


Danger Messages
Thoughts
Environment
Beliefs
Stress
Knowledge
Past experiences
Memories
Noise
Planning and predicting
Smell
Vision
Safety
Danger
PAIN
Inflammation
Temperature
Movement
Touch
Danger

Chronic Pain State

2. Cortical smudging:

- incorrect representation of painful body part
- precision lost: tactile, proprioception, movement, size/position perception etc
- reason pain spreads, can occur on other side, in adjacent joints etc
- As soon as 30 mins



Cognition and Pain

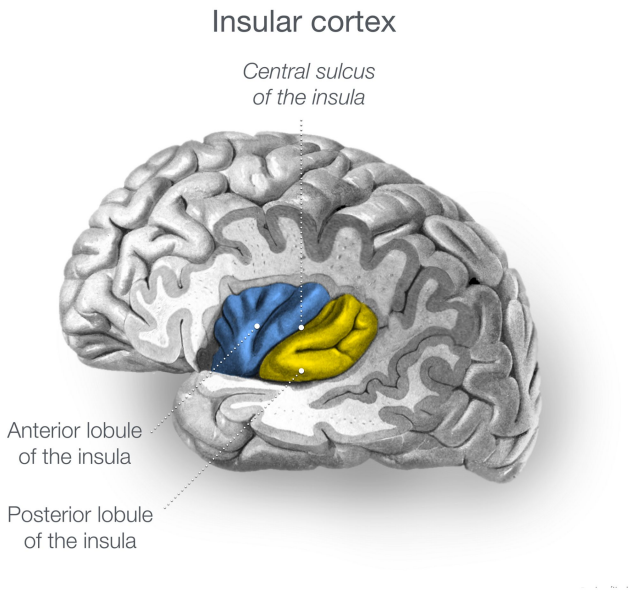
- To evaluate pain, we require the involvement of cognition
- Cognition involves the acquisition, processing, storage and retrieval of information by the brain
- A bi directional modulatory role between neural systems of cognition and pain

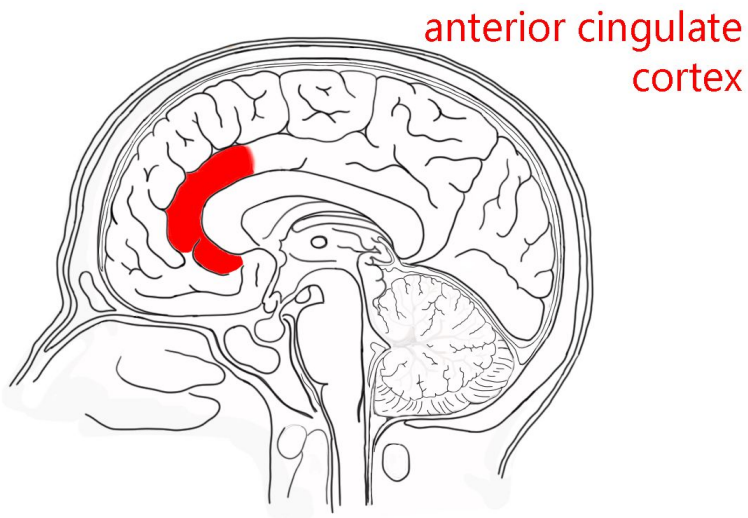
Cognition and Pain

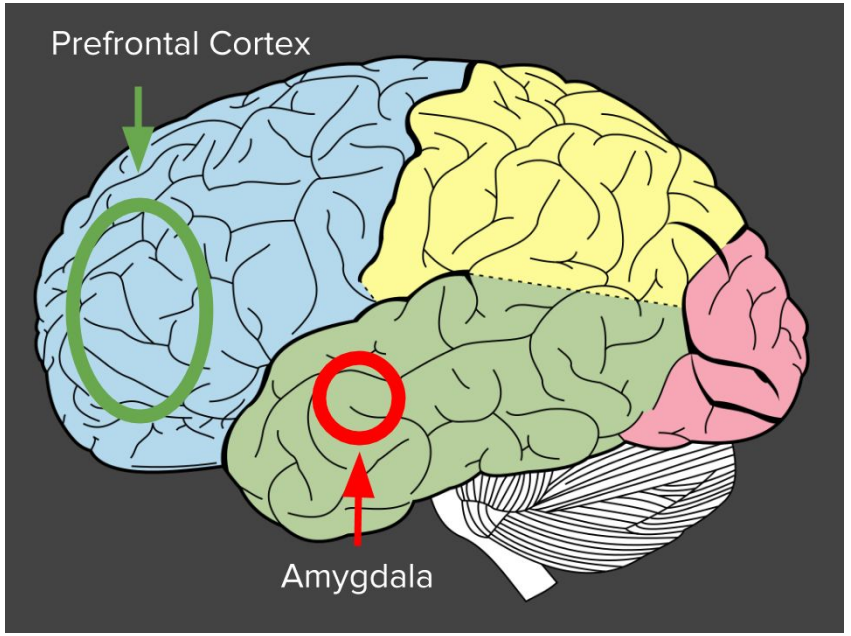
- **Attention:** Pain tends to capture attention, and individuals may focus more on pain, making the experience more intense
- **Memory:** The way we remember pain can also impact how we experience it in the future
- **Expectation:** The concept is connected to the placebo effect, where believing in the effectiveness of a treatment can reduce pain, even if no active treatment in provided

Emotions and Pain

- Pain and emotion are processed in similar brain areas
- Anterior cingulate cortex and the insula
- Chronic Pain and brain changes
- In areas related to emotional processing - such as the prefrontal cortex and amygdala

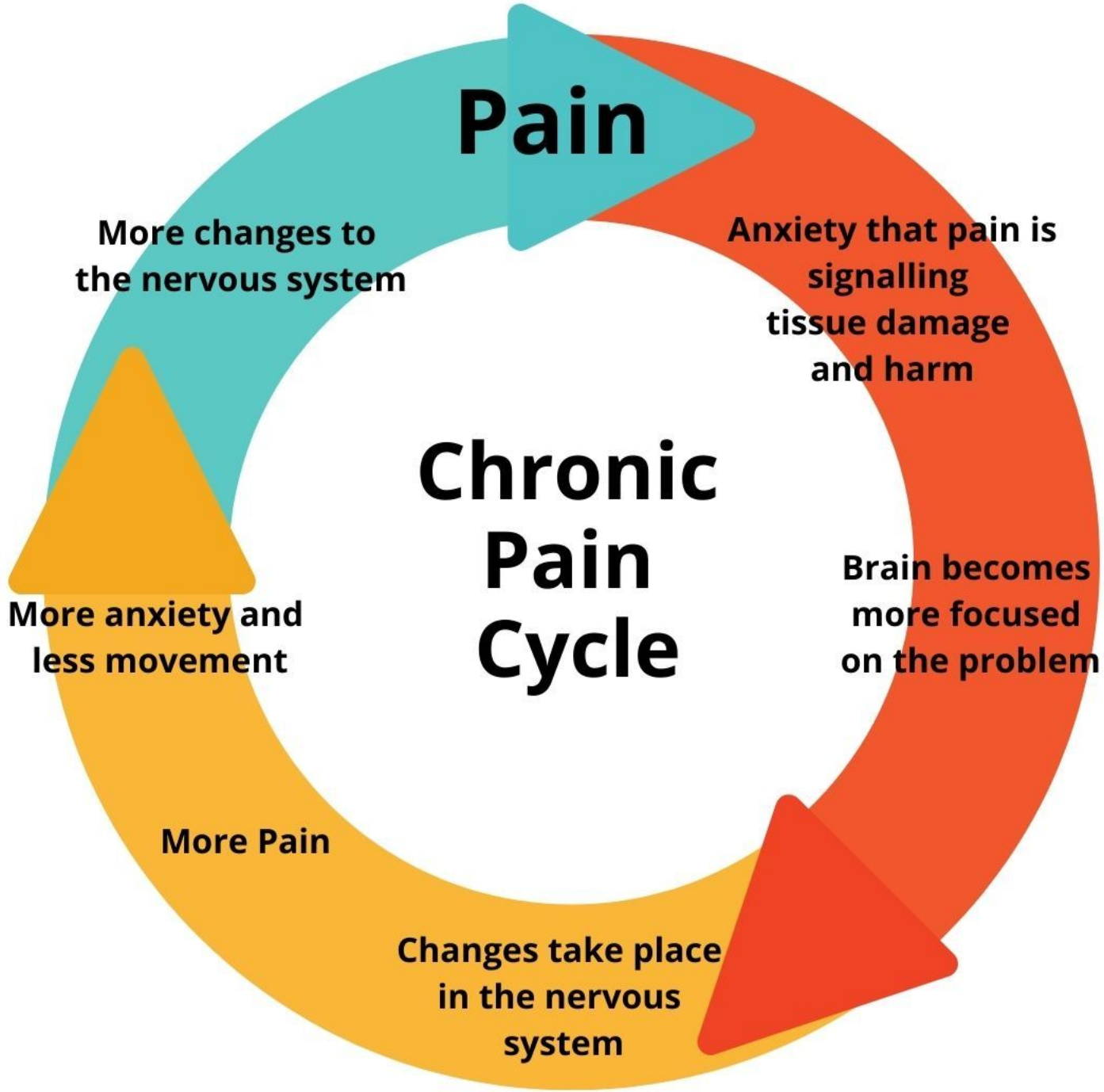






Emotional Dysregulation and Pain

- Increased emotional sensitivity
- Emotional blunting or numbing
- Increased anxiety and depression
- Frustration and irritability
- Fear and catastrophizing
- Stress response activation
- Chronic stress and emotional regulation
- Coping mechanisms
- Adaptive coping strategies
- Social isolation
- Interpersonal conflict


Pain
Anxiety that pain is signalling tissue damage and harm
Brain becomes more focused on the problem
Changes take place in the nervous system
More Pain
More anxiety and less movement
More changes to the nervous system
Chronic Pain Cycle

Our body is constantly changing and adapting to the information it receives everyday.

This is all reversible!

‘We need to understand why we hurt to stop hurting’

– David Butler

Resilience

- Goal of physiotherapy is a resilient system

  NOT pain-free system



Relevance

- Everything Matters in Pain
  - Stress
  - Sleep
  - Nutrition
  - Thoughts
  - Environment
- Everything Matters on Game Day
- Everything Matters in the Garden





Biopsychosocial Intervention

- A biopsychosocial intervention is defined as an intervention that includes a physical component (for example, exercise), and a psychosocial component (for example, awareness of cognitive influence on pain, self-management).

- Recommended by both national and international practice guidelines and systematic reviews.

Biomedical focus

- Take care to bend the knees when lifting.
- Use abdominal muscles to support the back.
- Biomechanics of the back.
- Description of exercises for back health
- Postural and mattress changes

(Roland and Dixon, 1989)

Education



1. **Listen: 2-3 sessions**

- Patient is the expert
- They already understand their pain

- Understand their beliefs and lives
- Find a point of relevance

De-constructing their beliefs and offering an alternative…

… best done by an analogy

Communication in pain management:

- Communication/education is a key component of pain management.
- Verbal/non-verbal
- Important to achieving desired goals/outcomes in pain management.
- Has a positive effect on self-empowerment, self-management, patients' knowledge and beliefs, engagement, perception, acceptance, motivation, and satisfaction with management received.

(Geneen et al., 2015; Wong et al., 2017)

Communication

- Need to get message across 7x to be retained
- Avg person takes in 10 000 pieces of information / day
- Standard of information we take in: average readability age of the guardian is 14 years
- We have 10 000 extra pieces of vocabulary

Simple, No health jargon, Analogies and Stories

Principles of communication in pain

- Physiotherapist – clear and simple, empathy, conscious of personal cues, managing expectations, collective action, reflexive monitoring
- Patient – trust, confidence, coherence, collective action, motivation, self-efficacy
- Validation and belief
- Open and non judgemental communication style
- Shared decision making
- Patient understanding
- Therapeutic alliance
- Patient centered approach

Resilient Neuro – Immune System

1. **Safe environment** —> Reassurance, listening
2. **Relevance** —> Understand their life, link their nervous and immune system to something they'll understand
3. **Gradual loading** —> they are current unable to process sensory & mechanical input appropriately
4. **De-construct poor pain beliefs** —> later, empower patient to do so, requires insight and energy

Key Points:

- Goal setting and pacing to encourage graded return to function.
- Encourage positive coping strategies.
- Discourage the premise “hurt=harm.”
- Specific advice on returning to daily activities
- Relapse is (almost) inevitable.
- May not be due to an individual event but multifactorial, challenging coping resources
- Written plans for managing acute episodes
- Make sure partner/family/carers are aware
- Explain how partner/family/carers can help
- Set specific targets to resume normal activities

Fear-avoidance model

Fear-Avoidance model (Vlaeyen, 2000), avoidance of feared movements as a consequence of catastrophic thinking.

Discuss and expose patients to feared movements, thus increasing physical function and reducing catastrophic thinking, directly measured through increase in previously feared movements and reduction in scores on the Pain Catastrophizing Scale.

Generalised improvement on disability and function scales, but not necessarily in pain levels.

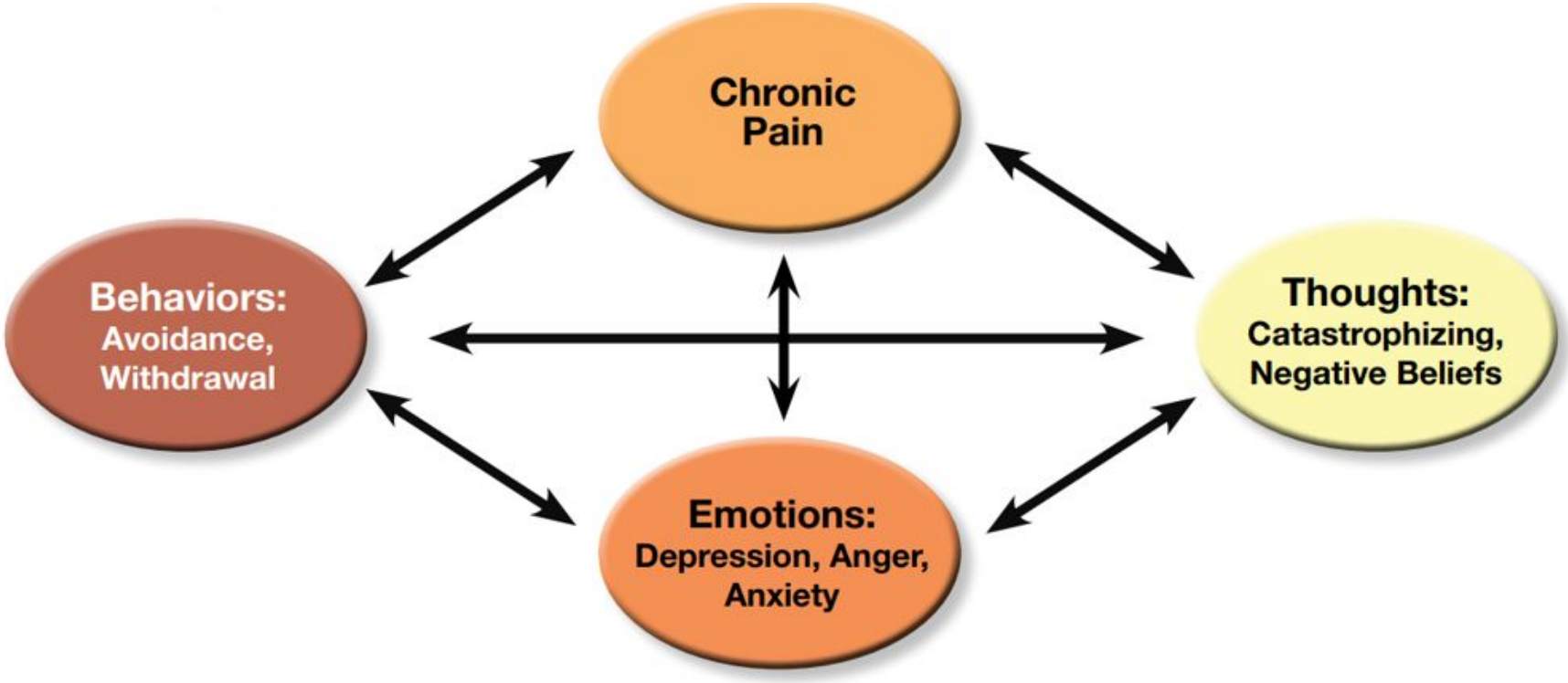
Cognitive behavioural therapy

- Change dysfunctional beliefs about pain.
- Reduce fear of injury – fear avoidance beliefs
- Increase self efficacy (confidence)
- Regulate activity (pacing)
- Achieve goals
- Reducing maladaptive behaviours.

(Hofmann et al., 2012)

CBT in Chronic Pain

- Cognitive restructuring
- Behaviour techniques
- Emotional regulation



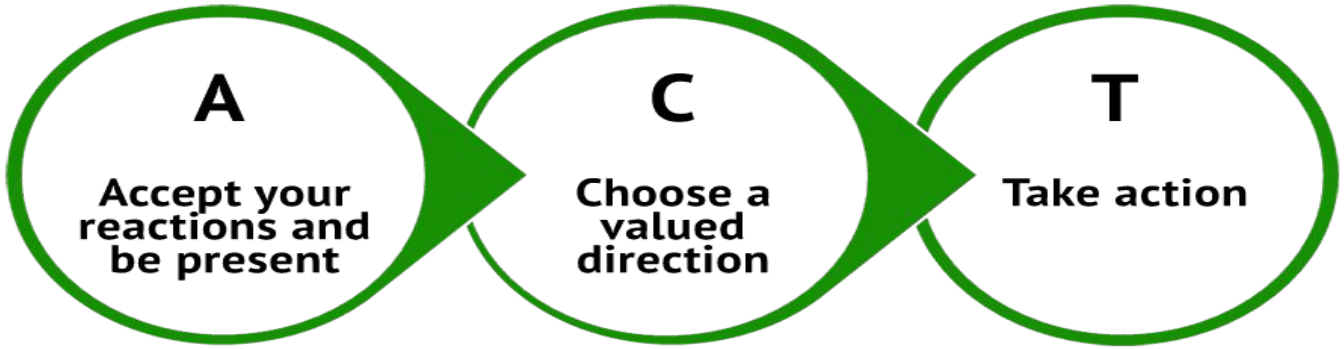
CBT Treatment for Chronic Pain

- The objectives are:

1. Reducing the negative impact of pain on daily life
2. Improving physical and emotional functioning
3. Increasing effective coping skills for managing pain
4. Reducing pain intensity

Acceptance and Commitment Therapy ACT

1. Developing acceptance of unwanted private experiences which are out of personal control
2. Commitment and action towards living a valued life
3. Type of psychotherapy that helps move beyond negative thoughts and feeling is a meaningful way
4. Encourages people to embrace challenging thoughts, feeling and memories rather than trying to change them
5. Focus on developing a more positive relationship with these experiences
6. ACT is effective for depression, anxiety, stress and chronic pain



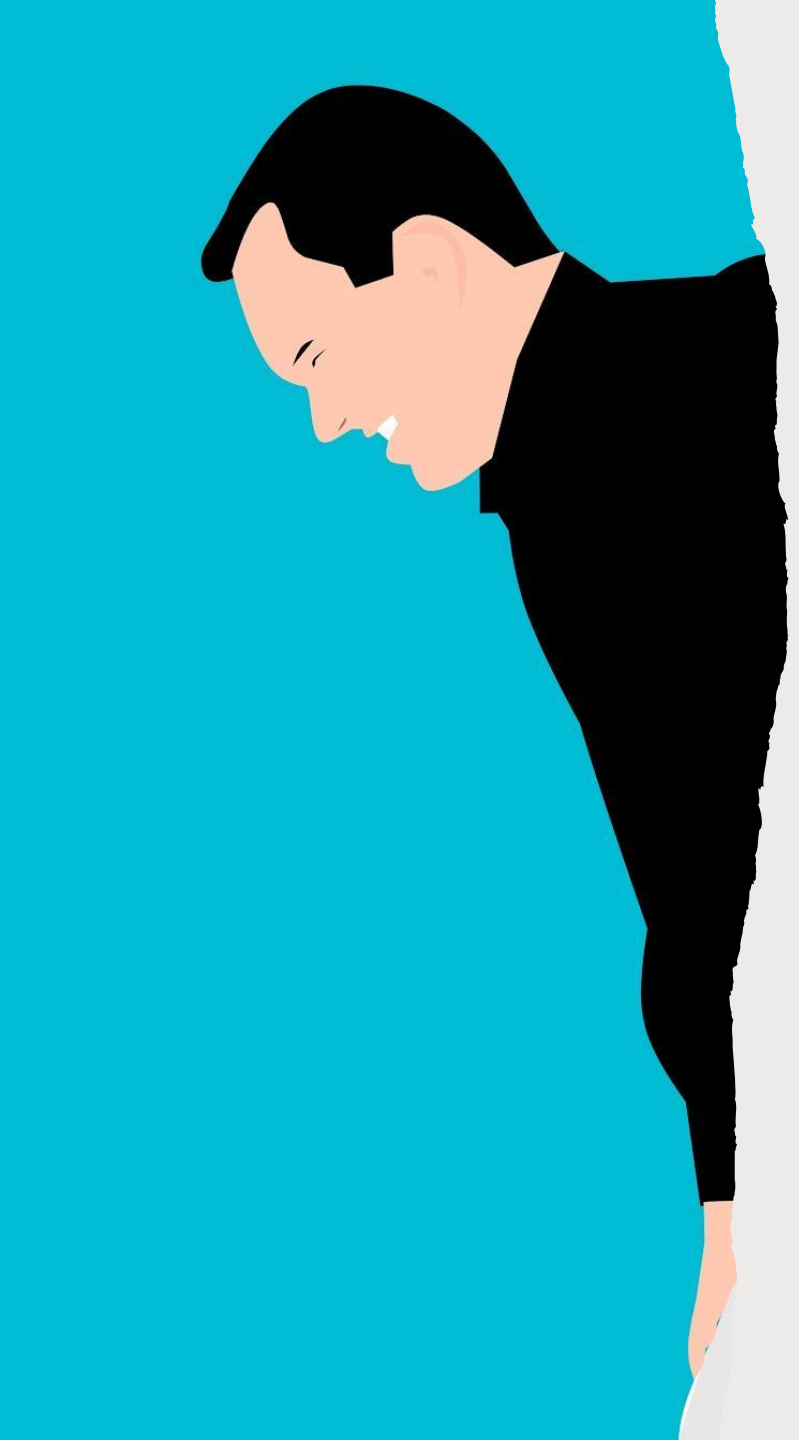
Key Principles:

1. **Acceptance:** opening up and making room for painful feelings, urges and sensations.
2. **Cognitive Defusion:** Detach from unhelpful thoughts, beliefs and memories
3. **Being in the Present Moment:** Engaging fully with your here-and-now experience with an attitude of openness, curiosity, interest and receptivity.
4. **The observing self/self as context:** being aware and can attend and detach from thoughts, feelings and actions.
5. **Values:** Understanding your core values
6. **Committed Action:** Creating goals with clear actionable steps that are aligned with your values.



Motivational interviewing

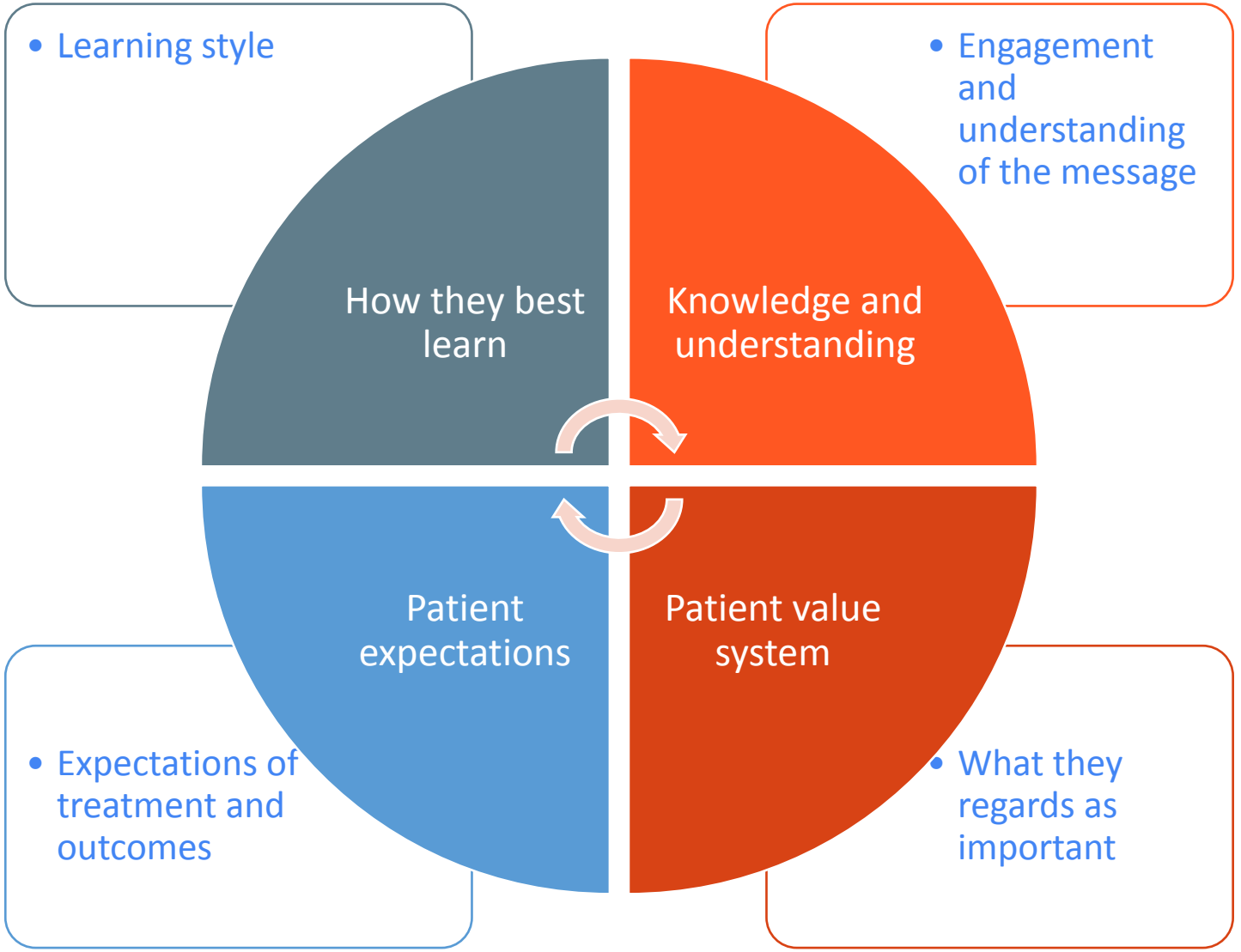
- “If you treat an individual as he is, he will stay as he is, but if you treat him as if he were what he ought to be and could be, he will become what he ought to be and could be
- MI is a communication-based intervention involving a patient-centred approach with the rationale to address patients’ uncertainty about behaviour change by reinforcing their own motivation and commitment to change (Miller and Rollnick, 2012). Clinically, MI adopts a transtheoretical model of behaviour change. That is:
  - Precontemplation
  - Contemplation
  - Preparation
  - Action
  - Maintenance



Pain neuroscience education

- Education/communication-based intervention which reduces pain and maladaptive beliefs and behaviours amongst patients (Moseley and Butler, 2015).
- Communication focused on reconceptualization of pain as “a marker of tissue damage” to a necessity of the body to protect itself from harm.
- Desensitising patients’ neural system neurobiology, neurophysiology, meaning of pain and how it is represented, instead of the traditional anatomical and biomedical model (for example, what is damaged and how can I repair it?).

Patient understanding



Shared decision making

1. Define/explain the issues, problem – teaching element
2. Present options and uncertainties – test understanding
3. Discuss pros and cons (uncertainty)
4. Explore patient preferences (values)
5. Discuss patient self-efficacy (implementation)

Patient-centred care

- A prerequisite for successful management.

- A moral philosophy of HCPs to buy into the delivery of high-quality healthcare.

- Characterised by eight indicators:

1. continuity;
2. respect for patient's preference, values, and needs;
3. education for the patient and carers;
4. emotional support;
5. access to care;
6. physical comfort;
7. participation of friends and family;
8. integrated and coordinated care.

Therapeutic alliance

- Achieved through positive communication from the physiotherapist.
- Patients report good communication and individualised care as facilitators to therapeutic alliance.
- TA – an intrinsic requirement to address patients' BPS factors.

Typical Words to Avoid and Alternatives for Patients	
Words to Avoid	**Alternatives**
Chronic degenerative changes	Normal age changes
Negative test results	Everything appears normal
Instability	Needs more strength and control
Wear and tear	Normal age changes
Neurological	Nervous system
Don't worry	Everything will be okay
Bone on bone	Narrowing/tightness
Tear	Pull

Damage	Reparable harm
Paresthesia	Altered sensations
Trapped nerve	Tight, but can be stretched
Lordosis	The normal curve in your back
Kyphosis	The normal curve in your back
Bulge/herniation	Bump/swelling
Disease	Condition
Effusion	Swelling
Chronic	It may persist, but you can overcome it
Diagnostics	X-ray or scan
You are going to have to live with this	You may need to make some adjustments

Helpful Facts

- Greatest improvement in chronic pain managed through education and gradual loading is seen between 6 and 12 months
- The immune system creates all change in the body – takes **up to** 2 weeks to do so

Why Physios

- Everyone we see is in acute or chronic pain
- Our response to acute pain will influence potential of chronic pain
- Use videos
- Prescribe something that won’t hurt —> Evidence for safety !